

Bachelor in Computer Applications

Course Title: **Network and Data Communications**

Code No: Semester: **4**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. What are the features of Distance Vector Routing protocol? Explain Distance Vector Routing protocol with relevant example. What are its disadvantages?
2. Explain the concept of Binary Phase Shift Keying. Represent bit sequence 111000110 by the following waveform. NRZ-lb. Differential Manchester
3. Explain the design of Selective Repeat ARQ. Illustrate it with suitable flow diagram example.
4. List different categories of networks and explain any two.
5. What is SNMP and how does it work?
6. What is MAC-address? The message sequence is 1011011 and generator polynomial $G(X) = x^3 + x^2 + 1$. Calculate the transmitted encoded frame.
7. How IGMP allows devices to join a multicast group? Explain.
8. Differentiate between reliable and unreliable protocol. Provide example of each. How can these protocols be used?
9. Explain different phases of packet switching. How packet switching works?
10. Explain Frequency Division Multiplexing with required figure.
11. A pure ALOHA network transmits 200-bit frames using a shared channel with a 200-kbps bandwidth. Find the throughput if the system considering all stations together produces 1000 frames per second. What do you mean by vulnerable time of pure ALOHA?
12. Write short notes on a. Shannon Capacity b. Connectionless service